

Chapter 06 Homework

Part A multiple choice (select the one that is best in each case. 1 point/question)

1. Determine which of the following statement is *true*.

- A) Electromagnetic radiation and sound waves travel at the same speed.
- B) Visible light is a form of electromagnetic radiation.
- C) Infrared light has shorter wavelengths than visible light.
- D) The frequency of radiation increases as the wavelength increases.
- E) Our bodies are penetrated by X rays but not by visible light. This is because X rays travel faster than visible light.

2. The Heisenberg uncertainty principle states that it is impossible to precisely know both the position and the _____ of an electron in an atom.

- A) color
- B) momentum
- C) mass
- D) shape
- E) velocity

3. If electrons are ejected from a given metal when irradiated with a 10-W red laser pointer, what will happen when the same metal is irradiated with a 5-W green laser pointer?

- A) Electrons will be ejected.
- B) Electrons will not be ejected.
- C) More information is needed to answer this question.

4. Consider the following three moving objects:

- (i) A golf ball with a mass of 45.9 g moving at a speed of 50.0 m/s.
- (ii) An electron with a mass of 9.11×10^{-28} g moving at a speed of 3.5×10^5 m/s.
- (iii) A neutron with a mass of 1.67×10^{-24} g moving at a speed of 2.3×10^2 m/s.

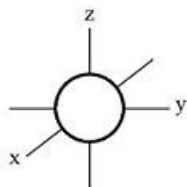
List the three objects in order from shortest to longest de Broglie wavelength.

- A) iii < i < ii
- B) i < ii < iii
- C) iii < ii < i
- D) ii < iii < i
- E) i < iii < ii

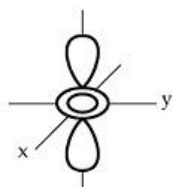
5. An orbital has $n = 4$ and $m_l = 0$. What are the possible values of l for this orbital?

- A) 0, 1, 2, 3
- B) -3, -2, -1, 0, 1, 2, 3
- C) 1, 2, 3
- D) -3, -2
- E) 1, 2, 3, 4

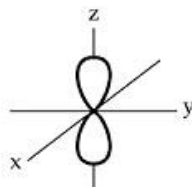
6. Of the *following* orbitals, _____ most resembles a p_x orbital.



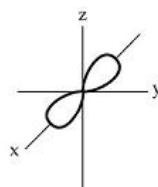
A)



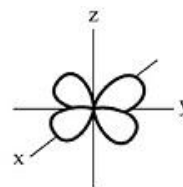
B)



C)



D)



E)

7. What is the wavelength of radiation that has a frequency of $5.11 \times 10^{11} \text{ s}^{-1}$?

- A) $5.87 \times 10^{-4} \text{ cm}$
- B) $5.87 \times 10^{-12} \text{ m}$
- C) $5.87 \times 10^{-2} \text{ m}$
- D) $5.87 \times 10^{-4} \text{ m}$

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8. The average distance from the nucleus of a $3s$ electron in a chlorine atom is smaller than that for a $3p$ electron. In light of this fact, which orbital is higher in energy?

- A) Not enough information
- B) These orbitals are equal in energy.
- C) The $3s$ is higher in energy than $3p$.
- D) The $3p$ is higher in energy than $3s$.

9. Which quantum numbers must be the same for the orbitals that they designate to be degenerate in a many-electron system?

- A) n , l , and m_l
- B) n only
- C) l and m_l
- D) m_l only
- E) n and l only

10. The lowest orbital energy is reached when the number of electrons with the same spin is maximized. This statement describes _____.

- A) Pauli Exclusion Principle
- B) Planck's constant
- C) Hund's rule
- D) Heisenberg Uncertainty Principle
- E) De Broglie hypothesis

11. Which of the following set of quantum numbers (ordered n , l , m_l , m_s) are possible for an electron in an atom?

- A) 5, 2, 1, $-1/2$
- B) 2, 2, 2, $1/2$
- C) 4, 3, 4, $-1/2$
- D) -3 , 2, 2, $-1/2$
- E) 3, 2, 1, -1

12. Which electron configuration represents a violation of the Pauli exclusion principle?

- A)

1s	2s	2p
↑↓	↑↓	↑↓
- B)

1s	2s	2p
↑↓	↑	
- C)

1s	2s	2p
↑↓	↑↑	↑
- D)

1s	2s	2p
↑↓	↑↓	↑ ↑
- E)

1s	2s	2p
↑	↑	↑ ↑ ↑

13. In the Bohr model of the atom, which hypothesis(es) is/are incorporated into quantum mechanics?

- A) electrons travel in circular paths called orbits
- B) electrons exist only in certain discrete energy levels
- C) energy is quantized in the transition of an electron from one level to another
- D) electron paths are controlled by probability

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E) both B) and C)

14. Which one of the following configurations depicts an excited oxygen atom?

A) $1s^2 2s^2 2p^2$

B) $1s^2 2s^2 2p^2 3s^2$

C) $1s^2 2s^2 2p^1$

D) $1s^2 2s^2 2p^4$

E) $[\text{He}] 2s^2 2p^4$

15. The photoelectric effect is _____.

A) the total reflection of light by metals giving them their typical luster

B) the production of current by silicon solar cells when exposed to sunlight

C) the ejection of electrons by a metal when struck with light of sufficient energy

D) the darkening of photographic film when exposed to an electric field

E) a relativistic effect

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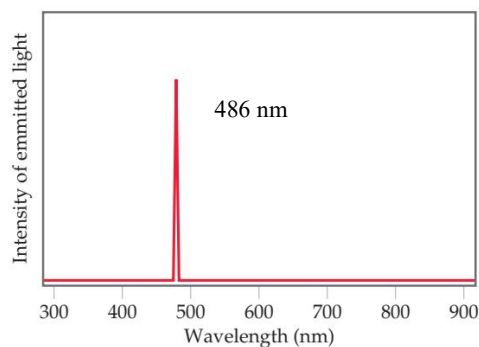
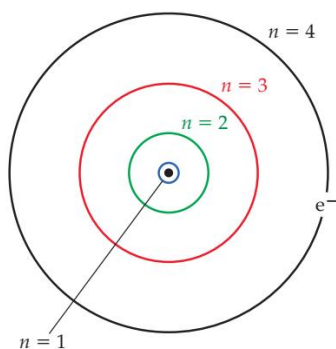
Part B short questions--Write legibly and show all work for all steps in the problem. For mathematical relationships show both the equations used and substitution of values into the equation. Maintain correct units throughout your calculations and report your answers with the correct significant figures. (15 points)

16. (6 points) In the Bohr model of the hydrogen atom, electrons are confined to orbits with fixed radii, and those radii can be calculated. The radii of the first four orbits are 0.53, 2.12, 4.76, and 8.46 Å, respectively, as depicted below.

(a) If an electron makes a transition from the $n_i = 4$ level to a lower-energy level, $n_f = 3, 2$, or 1, which transition would produce a photon with the shortest wavelength?

(b) What are the energy and wavelength of such a photon, and in which region of the electromagnetic spectrum does it lie?

(c) The image on the right shows the output of a detector that measures the intensity of light emitted from a sample of hydrogen atoms that have been excited so that each atom begins with an electron in the $n = 4$ state. What is the final state, n_f , of the transition being detected?



17. (3 points) Molybdenum metal must absorb radiation with a minimum frequency of $1.09 \times 10^{15} \text{ s}^{-1}$ before it can eject an electron from its surface via the photoelectric effect.

(a) What is the minimum energy needed to eject an electron?

(b) What wavelength of radiation will provide a photon of this energy?

(c) If molybdenum is irradiated with light of wavelength of 120. nm, what is the maximum possible kinetic energy of the emitted electrons?

18. (6 points) (a) Based on their position in the periodic table, write the **condensed electron configuration** for (1) F (element 9); (2) Zn (element 30); (3) Si (element 14).

(b) Indicate unpaired electrons of each atom in part (a).